



PRECISE SEISMIC CHARACTERISATION OF TESS LUMINARIES IN THE PLATO LOP FIELDS

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1 Main-sequence stars exhibit acoustic oscillations

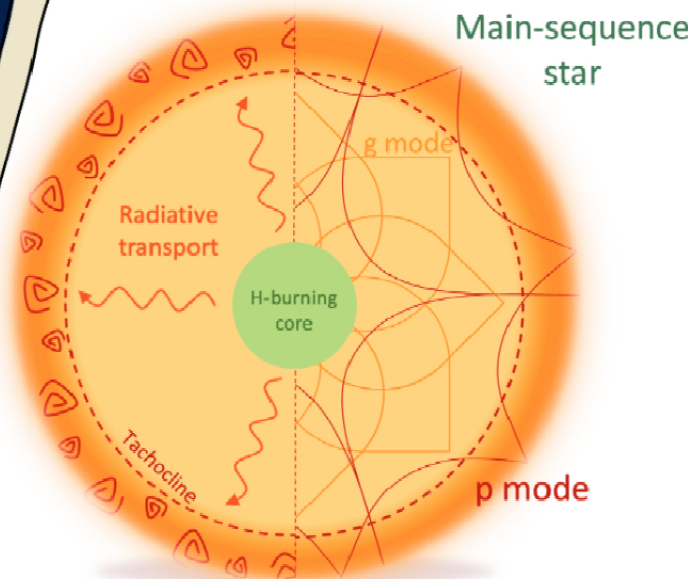
Watch! This star looks like the Sun. It naturally vibrates, just like a musical instrument.



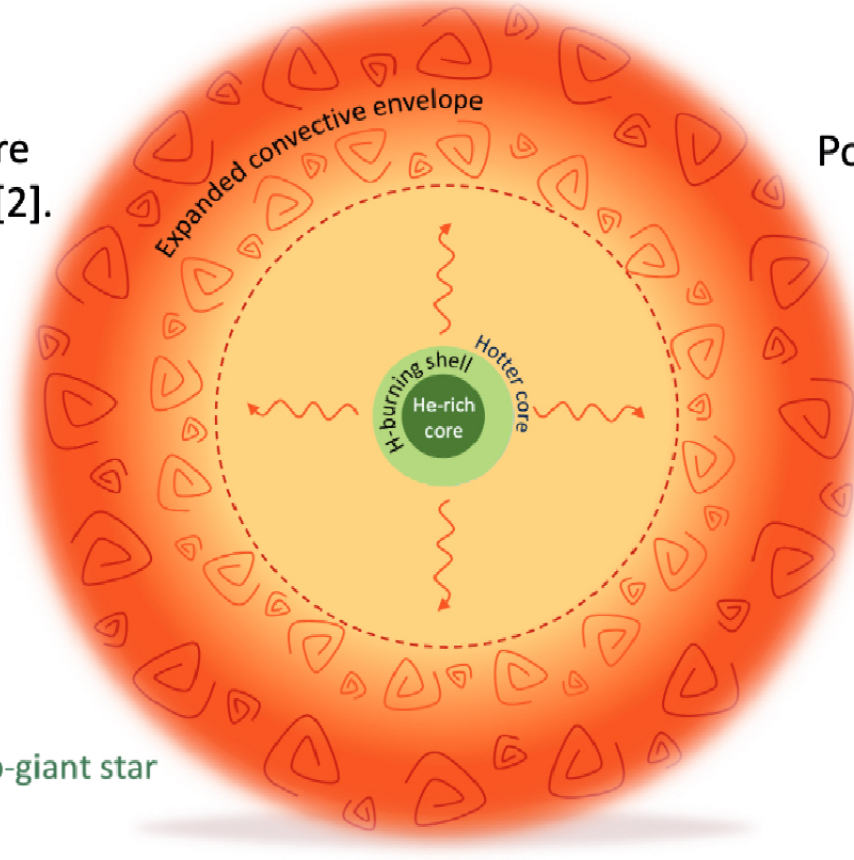
Its oscillation modes are visible in the photometric light curves. Their regular pattern is related to the radius, mass and internal properties of the star [1].

2 Then the star evolves...

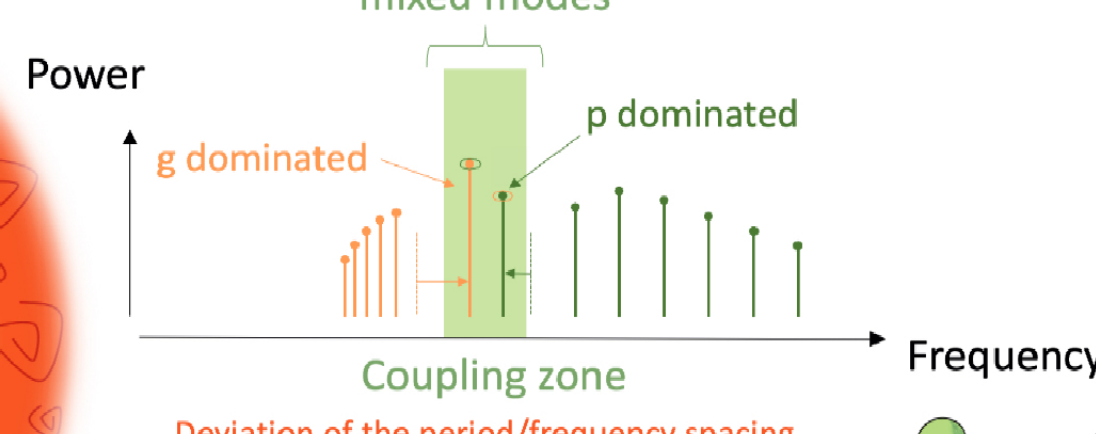
When the star leaves the main sequence, its core contracts and its convective envelope expands [2].



Sub-giant star



p- and g-mode cavities couple forming **mixed modes**



Awesome! They're probing **both the core and the surface** at the same time [3, 4].

3 Moreover, detecting and identifying the modes in hot main-sequence and sub-giant stars is not that easy...

The problem? Hotter stars have thinner convective zones, making their modes broader and harder to identify [5].

And that's not all! The mixed modes no longer follow the regular pattern of the p modes.

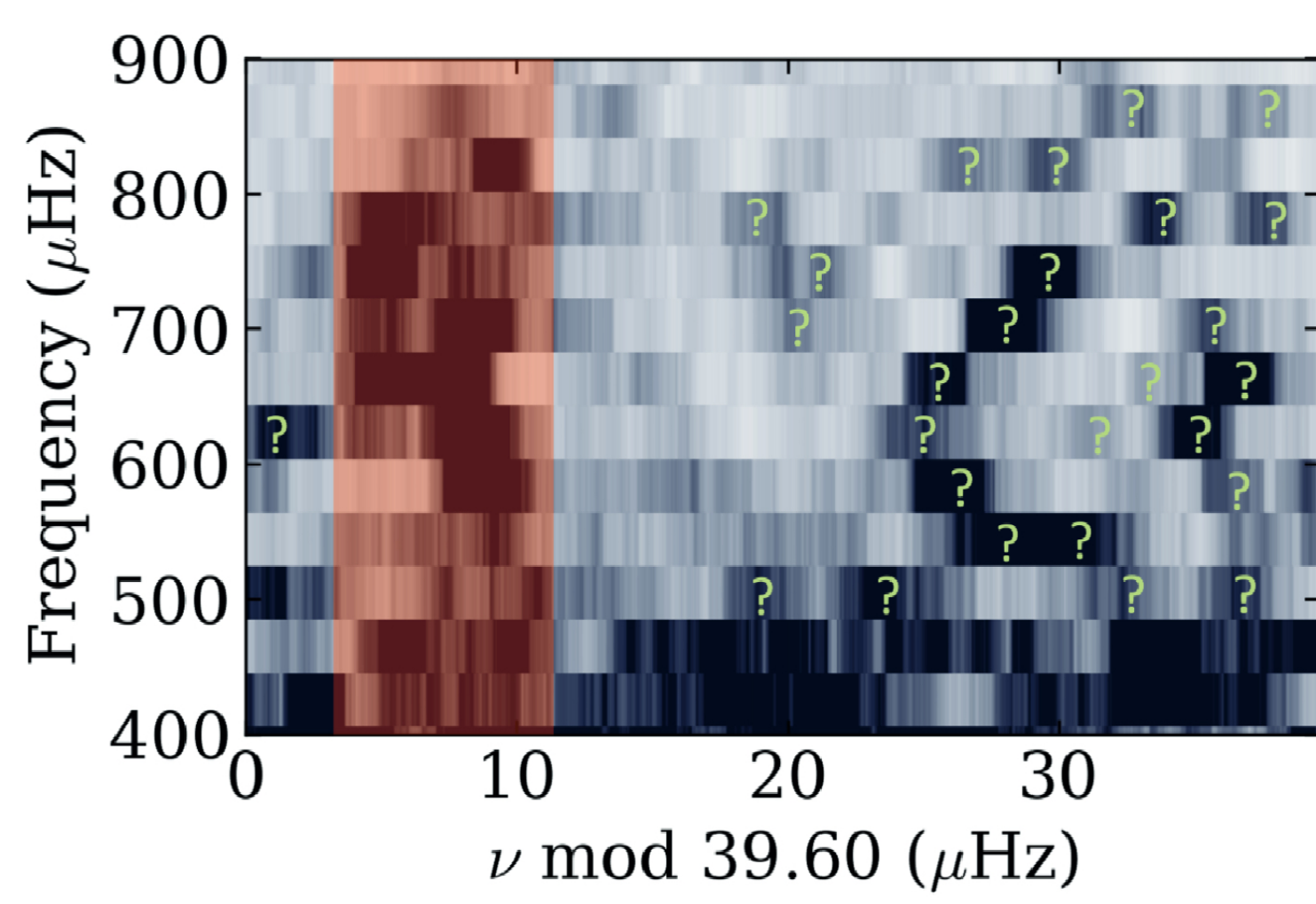
This is clearly visible in the échelle diagrams where mixed $\ell = 1$ modes deviate from the p-mode ridge [5, 6].



Modes in the red area should be associated with the $\ell = 0$ and $\ell = 2$ rigides

Is this one mixed?

This one too?



The first mixed modes arising during the sub-giant phase are due to a coupling of the p modes with non-asymptotic g modes. In this regime, **the asymptotic theory is no longer applicable**, preventing a simple analytical description of the mixed-mode frequencies.

4 And PLATO is coming!

PLATO will observe **tens of thousands of oscillating solar-like stars**, mainly in the main-sequence and sub-giant phases [7, 8].

It will be impossible to analyse everything manually! WE NEED AUTOMATION!

The reliability of the results must be tested and validated, in the early stages of the mission.

5 Our approach: Extracting the precise mode properties of 32 bright main sequence and sub-giant stars from the TESS Luminaries sample that will be observed by PLATO [9, 10]...

Used in its manual mode, with visually estimated guesses and inspection of the results [11]

apollinaire

FAMED PIPELINE

Data-driven approach based on the signal-to-noise significance + Bayesian model comparison [12, 13]

PBjam

The basis of the official PLATO pipeline
Physics informed guesses from empirical relations + thousands of observed and modelled stars [14, 15]

I've invited my friends to apply their preferred peak-bagging pipeline to the TESS observations of these stars.

6 Key results



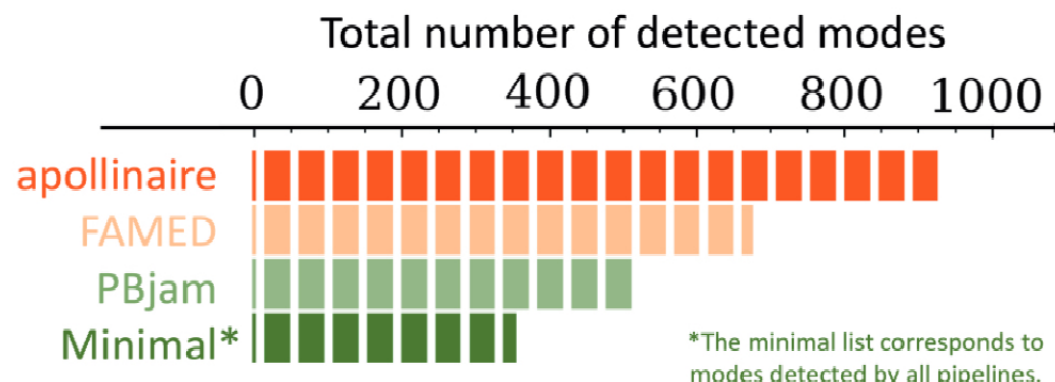
The identification of $\ell = 0$ and $\ell = 1$ modes is not consistent across pipelines
→ More significant in hotter stars: both identifications are consistent with the $\varepsilon - T_{\text{eff}}$ relation [16]
→ Need for a way to distinguish the best identification automatically in this temperature regime



For main sequence stars, extracted mode parameters are in very good agreement between the pipelines

→ But there are differences in the number of detected modes for all stars

The modes extracted simultaneously by the three pipelines are robust mode detections!



For subgiant stars, the pipelines do not detect the same mixed modes

29

fully characterised stars

26

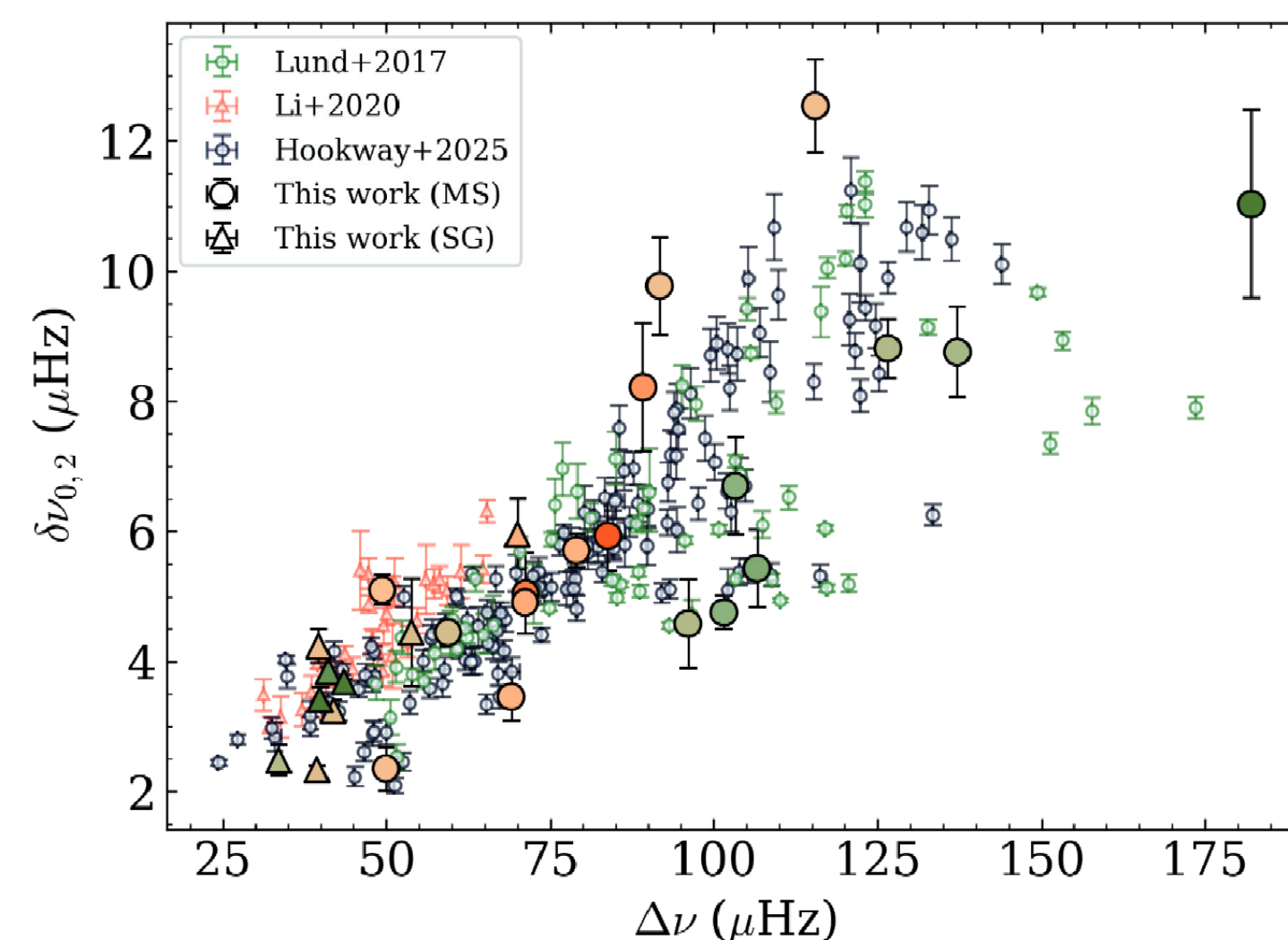
for the first time

A consistent and optimal reference catalogue for comparing future seismic measurements from PLATO (to be modelled in Bétrisey et al. in prep.)



TOWARD FULLY AUTOMATED MODE IDENTIFICATION IN HOT STARS AND SUBGIANTS!

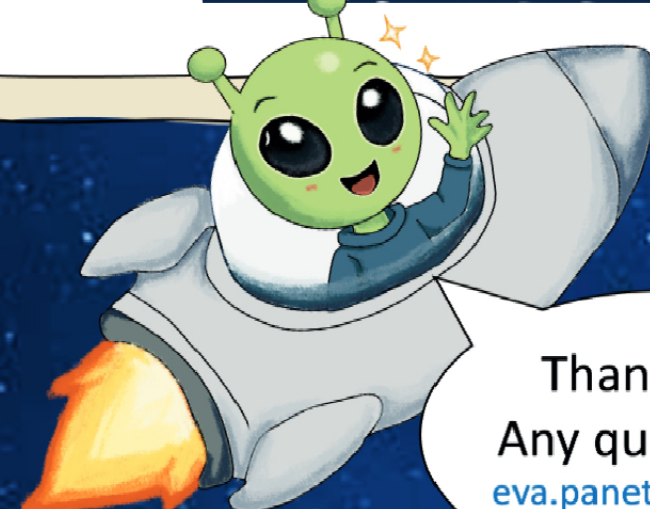
7 Let's compare the results with similar Kepler and K2 stars!



6750
6500
6250
6000
5750
5500
 T_{eff} (K)

Large and small separations of stars from our sample agree well with literature Kepler and K2 targets [17, 18, 19].

More in the paper here



Thank you!
Any questions?
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